

Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
DEPARTMENT OF COMPUTER ENGINEERING

Third Year (2019 Course)

310244: Computer Networks and Security
[Semester- V]

Unit – II
Data Link Layer

Course In-charge: Ms. Pooja M. Hande

Teaching Scheme:	Credit: 03	Examination Scheme:
TH: 03		Mid-Sem (TH) : 30 Marks
Hours/Week		End-Sem (TH): 70 Marks

Prerequisites Courses: --

Companion Course: Computer Networks and Security Laboratory (310247)

Course Objectives:

- To understand the fundamental concepts of networking standards, protocols and technologies
- To learn different techniques for framing, error control, flow control and routing
- To learn different layer protocols in the protocol stacks
- To understand modern network architectures with respect to design and performance
- To learn the fundamental concepts of Information Security

Course Outcomes:

On completion of the course, learners should be able to

CO1: Summarize fundamental concepts of Computer Networks, architectures, protocols and technologies

CO2: Illustrate the working and functions of data link layer

CO3: Analyze the working of different routing protocols and mechanisms

CO4: Implement client-server applications using sockets

CO5: Illustrate role of application layer with its protocols, client-server architectures

CO6: Comprehend the basics of Network Security

Reference & Text Books

Learning Resources
Text Books : 1. Fourauzan B.,"Data Communications and Networking", 5th Edition, Tata McGraw-Hill, Publications, ISBN:0–07 – 058408 – 7 2. Andrew S. Tanenbaum, Computer Networks, 5th Edition, Pearson India, 2012.
Reference Books : 1. Kurose, Ross, "Computer Networking a Top Down Approach Featuring the Internet", Pearson, ISBN-10: 0132856204 2. L. Peterson and B. Davie, "Computer Networks: A Systems Approach", 5th Edition, Morgan-Kaufmann, 2012. 3. Douglas E. Comer & M.S Narayanan,"Computer Network & Internet", Pearson Education 4. William Stallings, "Cryptography and Network Security: Principles and Practice", 4th Edition 5. Pachghare V. K., "Cryptography and Information Security", 3rd Edition, PHI,

Course Contents

Unit II	Data Link Layer	08Hours
Introduction, functions**. Design Issues:** Services to Network Layer, Framing. **ARQ strategies:** Error Detection and correction, Parity Bits, Hamming Codes (11/12-bits) and CRC. **Flow Control Protocols:** Unrestricted Simplex, Stop and Wait, Sliding Window Protocol. **WAN Connectivity:** PPP and HDLC. **MAC Sub layer:** Multiple Access Protocols: Pure and Slotted ALOHA, CSMA, WDMA, CSMA/CD, CSMA/CA, Binary Exponential Back-off algorithm, Introduction to Ethernet IEEE 802.3, IEEE 802.11 a/b/g/n, IEEE 802.15 and IEEE 802.16 Standards.		

Introduction

- Data Link layer is the **second layer of the OSI model**
- Data link layer responsible for converting data stream to signal bit by bit and send to send that over the underlying hardware. At the receiving end data link layer picks up the data from hardware which are in the form of electrical signals, assembles them in a recognizable frame format and hands over to the upper layer.
- **Data link layer has two sub layer:**
- **1. LLC(Logical Link Control Layer)-** It deals with protocols ,flow control and error control.
- **2.MAC(Media Access Control)-**It deals with actual control of media.
- **Devices Operating at the Data Link Layer-** Switch, Bridge, NIC,WAP,Layer 2 Switch

Functions of Data Link Layer

- **1. Framing-** Data Link Layer takes the packet from the network layer and encapsulate them into frames . Then, it sends each frame bit by bit on the hardware . At the receiver end data link layer picks up signals from hardware and assembles them into frames.
- **2.Addressing-** Data link provide Layer-2 hardware addressing mechanism. Hardware address is assumed to be unique on the link . It encoded into hardware at the time of manufacturing.
- **3. Synchronization-** When data frames are sent on the link , both machines must be synchronized in order to transfer take place.
- **4.Error Control-** Some time signals may have encountered problems in transition and bit are flipped . These errors are detected and attempted to recover actual data bits. It also provide error reporting mechanism to sender.

Functions of Data Link Layer

- **5. Flow Control -** Stations on same link may have different speed or capacity . Data link layer ensure flow control that enables both machine to exchange data on same speed.
- **6.Multi-Access –** When host on shared link tries to transfer the data . It has high probabilities of collision . Data link layer provides mechanism such as CSMA/CD to equip capabilities of accessing a shared media among multiple system.

Data Link Layer Design Issues-

- Data link layer takes the packets from network layer and encapsulate them into frames for transmission on physical layer.
- The frame contains a frame header, a payload field for holding the packet, and a frame trailer. as illustrated in figure below.

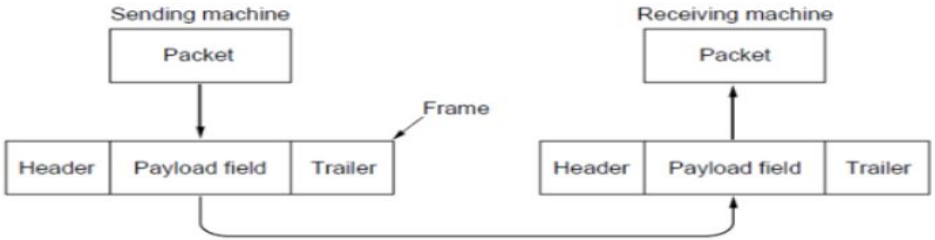


Figure : Relationship between packets and frames.

Data Link Layer Design Issues-

Services Provided to the Network Layer

- The function of the data link layer is to provide services to the network layer. The principle service is transferring data from the network layer on source machine to the network layer on destination machine.
- The data link layer can be designed to offer various services vary from protocol to protocol as mentioned below.
- 1. Unacknowledged connectionless service.
- 2. Acknowledged connectionless service.
- 3. Acknowledged connection-oriented service.

Data Link Layer Design Issues-

- **Unacknowledged connectionless service:**
- There is No logical connection establishment at beginning and released afterwards.
- The source machine send independent frames to the destination machine without having the destination machine acknowledge them.
- If a frame is lost due to noise, no attempt is made to detect or recover.
- This class of service is appropriate when error rate is very low and error recovery is left in higher layers.
- This types of service is used for real time traffic, such as voice, in which late data is not allowed.
- The Example is Ethernet.

Data Link Layer Design Issues-

- **Acknowledged connectionless service:**
- There is no logical connections used
- Each frame sent is individually acknowledged by receiver, by this way sender knows whether a frame arrived correctly or been lost. If it has not arrived within a specified time interval. It can be sent again.
- This service is useful over unreliable channels
- Example is Wireless system.

Data Link Layer Design Issues-

- **Acknowledged connection-oriented service.**
- This is most sophisticated service the data link layer can provide to the network layer .
- The source and destination machines establish a connection before any data are transferred.
- In Connection oriented service the data transfer go through three distinct phases.
    - Phase 1: Connection Establishment by having both sides initialize variables and counters needed to keep track of which frames have been received and which one have not.
    - Phase 2: The Frames are transmitted .
    - Phase 3: The Connection is released, freeing up the variables, buffers, and other resources used to maintain the connection.
- Each frame sent over the connection is numbered.
- Data link layer guarantees that each frame sent is indeed received and each frame is received exactly once and all frames are received in the right order.
- It is appropriate over long, unreliable links such as a satellite channel or long-distance telephone circuits.

Data Link Layer Design Issues-

- **Framing-**
- The physical layer accepts raw bit stream and attempt to deliver it the destination. If the channel is noisy, the bit received by data link layer is not guaranteed to be error free. Some bits may have different values and number of bits received may be in correct. It is up to the data link layer to detect and correct the errors.
- The data link layer break up the bit stream into discrete frames, compute a short token called a checksum for each frame, and include the checksum in the frame when it is transmitted . When a frame arrives at the destination, the checksum is recomputed. If the newly computed checksum is different from the one contained in the frame, the data link layer knows that an error has occurred and takes steps to deal with it. (e.g., discarding the bad frame and possibly sending back the error report).

Framing

A frame has the following parts

- **Frame Header** It contains the source and the destination addresses of the frame.
- **Payload field** It contains the message to be delivered.
- **Trailer** It contains the error detection and error correction bits.
- **Flag** It marks the beginning and end of the frame.

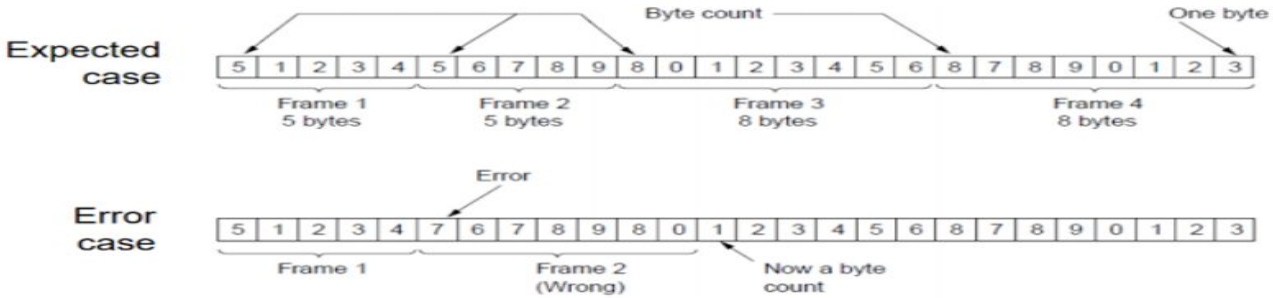


Framing

- Types of Framing
- Framing can be of two types, fixed sized framing and variable sized framing.
- **Fixed-sized Framing**
- Here the size of the frame is fixed and so the frame length acts as delimiter of the frame. Consequently, it does not require additional boundary bits to identify the start and end of the frame.
- Example ATM cells.
- **Variable Sized Framing**
- Here, the size of each frame to be transmitted may be different. So additional mechanisms are kept to mark the end of one frame and the beginning of the next frame.
- It is used in local area networks.

Framing-

- **The following framing methods were used by data link layer to form the frames.**
- Byte Count
- Flag bytes with byte stuffing
- Flag bits with bit stuffing
- **1. Byte Count :**
- In this method the first field in the header to specify the number of bytes in the frame. The data link layer at receiver sees the byte count, it knows how many bytes follow and where to end the frame.



Framing

- **2. Flag bytes with byte stuffing**

  In this method each frame start and end with special bytes called a flag byte. Two consecutive flag bytes indicate the end of one frame and the start of the next frame. Thus if the receiver ever loses synchronization it can just search for two flag bytes to find the end of the current frame and the start of the next frame.

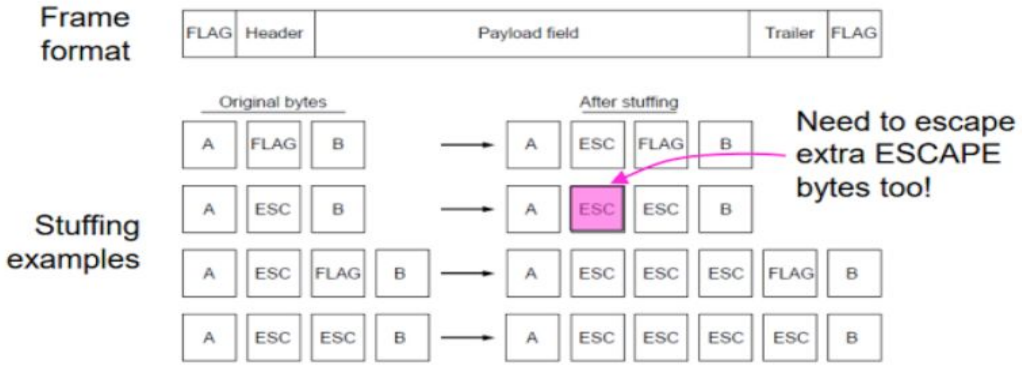


Figure : a) A frame delimited by flag bytes. b) Four examples of byte sequences before and after stuffing.

Framing-

- **3. Flag bits with bit stuffing.**
- In this algorithm each frame begins and ends with a special bit pattern(flag byte), 01111110 (0x7E in hexadecimal).
- Whenever the senders data link layer encounters five consecutive 1s in the data, it automatically stuffs a 0 bit into the outgoing bit stream to distinguish from flag byte. In the receiver sees five consecutive incoming 1 bits, followed by a 0 bit, it automatically destuffs (i.e., delete) the 0 bit. The following figures gives example.

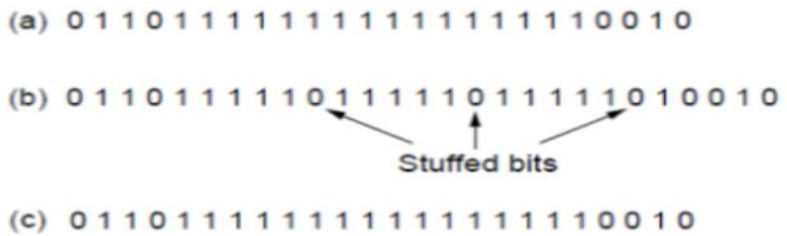


Figure: Bit Stuffing. a) The original data. b) The data as they appear on the line, c)The data as they are stored in the receivers memory after destuffing.

Error Control

- The protocol calls for the receiver to send back special control frames bearing positive or negative acknowledgements about the incoming frames. If the sender receives a positive acknowledgement about a frame, it knows the frame has arrived safely. On the other hand, a negative acknowledgement means that something has gone wrong and the frame must be transmitted again.
- The possibility that hardware troubles may cause a frame to vanish completely. In this case, the receiver will not react at all, since it has no reason to react and acknowledgement frame is lost, the sender will not know how to proceed. It should be clear that protocol in which the sender transmit a frame if a frame is ever lost due to malfunctioning hardware or faulty communication channel.
- The above situation is dealt with by introducing timers in the data link layer . When the sender transmits a frame, it starts a timer. The timer is set to expire after an interval long enough for the frame to reach the destination, and have acknowledgement propagate back to the sender. The frame will be correctly received and the acknowledgment will get back before the timer runs out.
- However, if either the frame or the acknowledgement is lost, the timer will go off, alerting the sender to a potential problem. The sender transmit the frame again. However, when frames may be transmitted multiple times there is a possibility of receiver will accept the same frame two or more time and send it to network layer more than once. To prevent this from happening, each outgoing frames are assign a sequence number, so that receiver can distinguish retransmission from originals.
- The timers and sequence numbers are used to ensure each frame is passed to the network layer at the destination exactly once. It is an important duty of data link layer.

Flow Control

- This is an important design issue where the sender is running on a fast powerful computer and receiver is running on slow, low-end machine. The receiver may be unable to handle the frames as fast as they are arrive and will lose some.
- Two approaches are commonly used for the above issue.
- 1. Feedback-based flow control.
- 2. Rate-based flow control.
- In **feedback-based flow** control, the receiver sends back information to the sender giving it permission to send more data or at least telling the sender how the receiver is doing.
- The feedback-based flow control is used in higher layers too.
- In **rate-based flow** control, the protocol has a built-in mechanism that limits the rate at which senders may transmit data, without using feedback from the receiver.

ARQ strategies: Error Detection & Correction

ARQ strategies: Error Detection & Correction

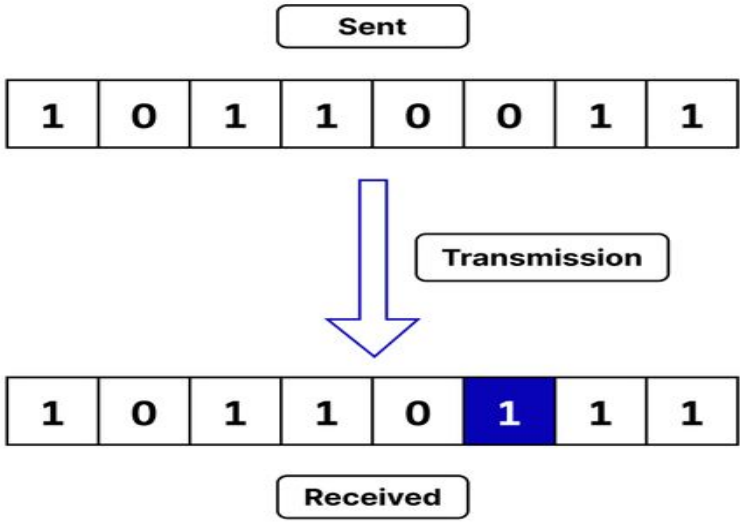
- **ARQ** stands for **Automatic Repeat Request** also known as **Automatic Repeat Query**. ARQ is an error-control strategy used in a two-way communication system. It is a group of error-control protocols to achieve reliable data transmission over an unreliable source or service.
- In these protocols, the receiver sends an acknowledgement message back to the sender if it receives a frame correctly. If the sender does not receive the acknowledgement of a transmitted frame before a specified period of time, i.e. a timeout occurs, the sender understands that the frame has been corrupted or lost during transit. So, the sender retransmits the frame. This process is repeated until the correct frame is transmitted.

Error Detection

- Error is a condition when the receiver's information does not match the sender's. Digital signals suffer from noise during transmission that can introduce errors in the binary bits traveling from sender to receiver. That means a 0 bit may change to 1 or a 1 bit may change to 0.
- Data (Implemented either at the Data link layer or Transport Layer of the OSI Model) may get scrambled by noise or get corrupted whenever a message is transmitted. To prevent such errors, error-detection codes are added as extra data to digital messages. This helps in detecting any errors that may have occurred during message transmission.

Error Detection

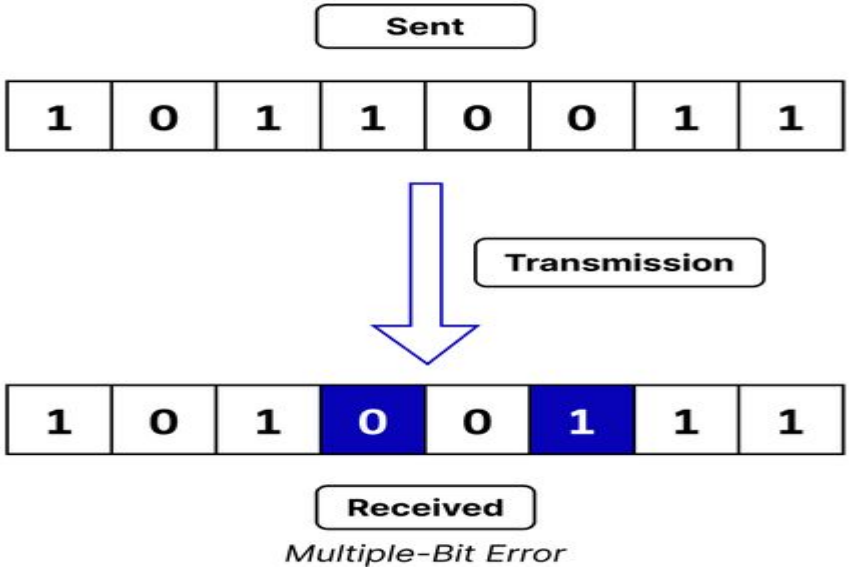
- **Types of Errors**
- **1. Single-Bit Error:** This type of error occurs when one bit of a transmitted data unit is altered, leading to corrupted data.



Single-Bit Error

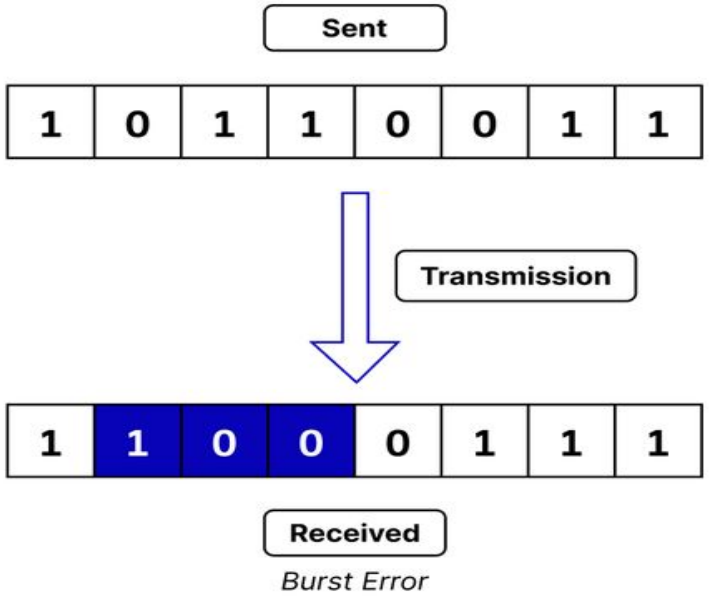
Error Detection

- **Types of Errors-**
- **2. Multiple-Bit Error:** This type of error occurs when more than one bit is affected. While rarer than single-bit errors, they can occur in high-noise environments.



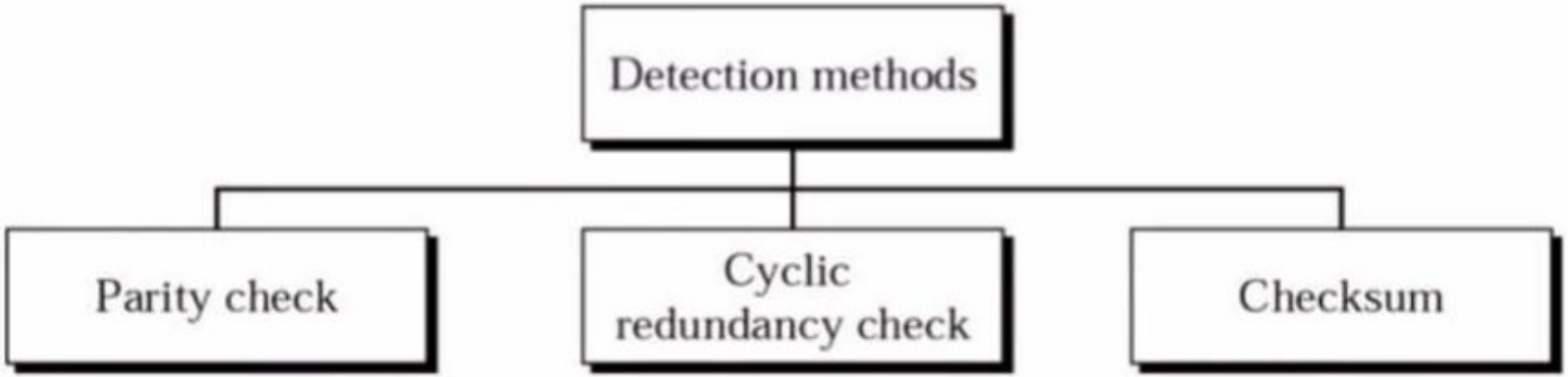
Error Detection

- **Types of Errors-**
- **3. Burst Error:** This type of error occurs when a sequence of consecutive bits is flipped, resulting in several adjacent bits being incorrect.



Burst Error

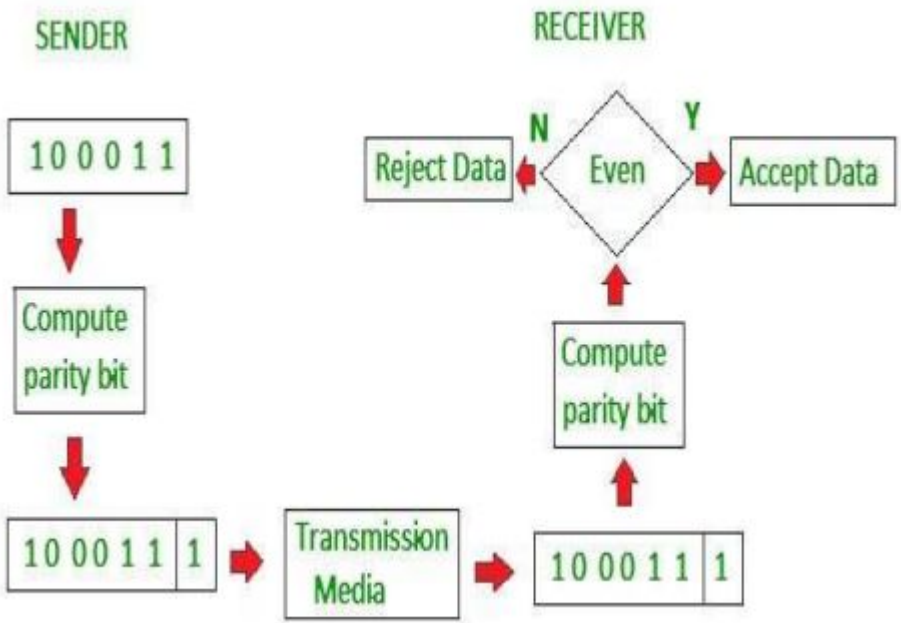
Error Detection Techniques



Error Detection Techniques

- 1. **Parity Bits**
- Parity is a simple error detection method that involves adding an extra bit to a data transmission. It works as:
- 1 is added to the block if it contains an odd number of 1's, and
- 0 is added if it contains an even number of 1's
- This scheme makes the total number of 1's even, that is why it is called even parity checking.

Error Detection Techniques



Error Detection Techniques

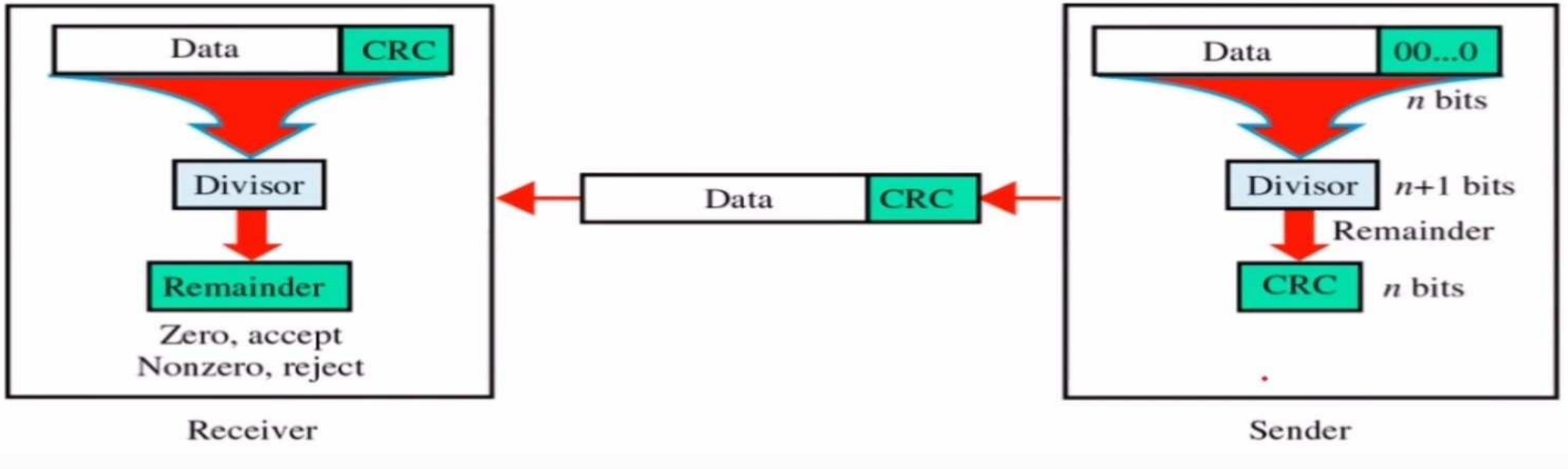
- **2.Cyclic Redundancy Check**
- Cyclic Redundancy Check (CRC) involves binary division of the data bits being sent by a predetermined divisor agreed upon by the communicating system. The divisor is generated using polynomials.
- Here, the sender performs binary division of the data segment by the divisor. It then appends the remainder called CRC bits to the end of the data segment. This makes the resulting data unit exactly divisible by the divisor.
- The receiver divides the incoming data unit by the divisor. If there is no remainder, the data unit is assumed to be correct and is accepted. Otherwise, it is understood that the data is corrupted and is therefore rejected.
- A form of channel coding known as a Cyclic Redundancy Code (CRC) is used in high-speed data networks.

Error Detection Techniques

- **2.Cyclic Redundancy Check**

❑ **CRC(Cyclic Redundancy Check)**

~ is based on binary division.

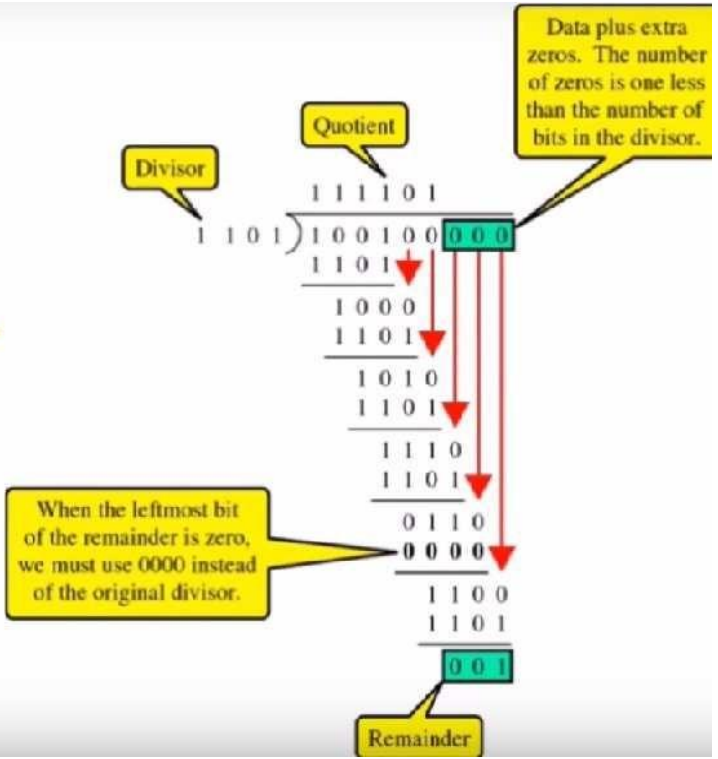


Error Detection Techniques

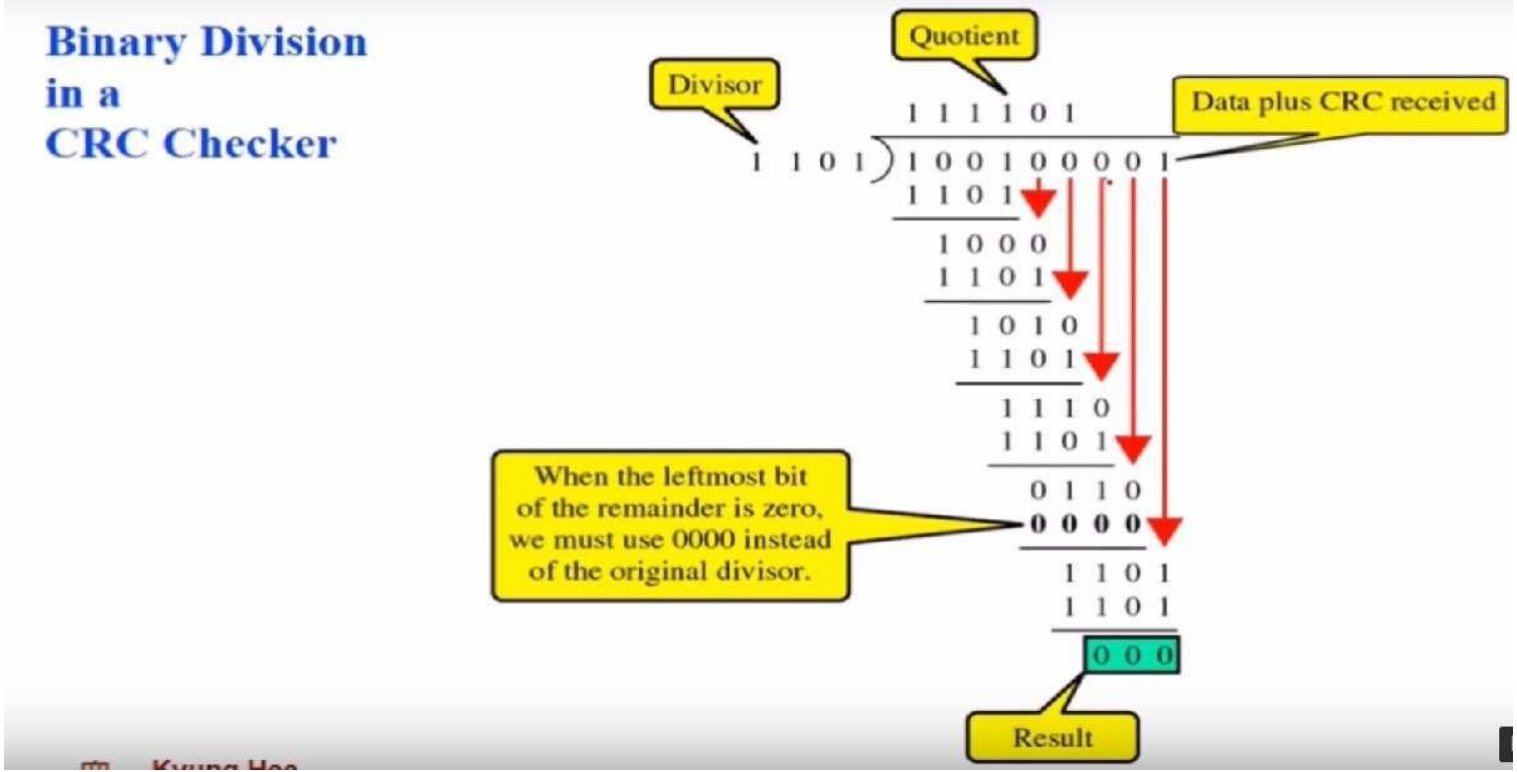
- CRC generator

~ uses modular-2 division.

Binary Division in a CRC Generator



Error Detection Techniques CRC

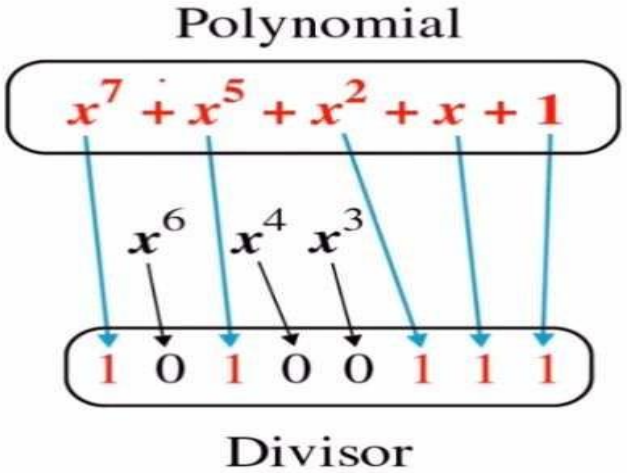


Error Detection Techniques CRC

- Polynomials
  - CRC generator(divisor) is most often represented not as a string of 1s and 0s, but as an algebraic polynomial.

$$x^7 + x^5 + x^2 + x + 1$$

- A polynomial representing a divisor



original message

1 0 1 0 0 0 0

@ means X-OR

Generator polynomial

x^3+1

$1.x^3+0.x^2+0.x^1+1.x^0$

CRC generator

1 0 0 1 4-bit

If CRC generator is of n bit then append (n-1) zeros in the end of original message

Sender

```
1 0 0 1 | 1 0 1 0 0 0 0 0 0 0
        @ 1 0 0 1
          ---------
          0 0 1 1 0 0 0 0 0 0
            @ 1 0 0 1
              ---------
              0 1 0 1 0 0 0 0
                @ 1 0 0 1
                  ---------
                  0 0 1 1 0 0 0
                    @ 1 0 0 1
                      ---------
                      0 1 0 1 0
                        @ 1 0 0 1
                          ---------
                          0 0 1 1
```

Message to be transmitted

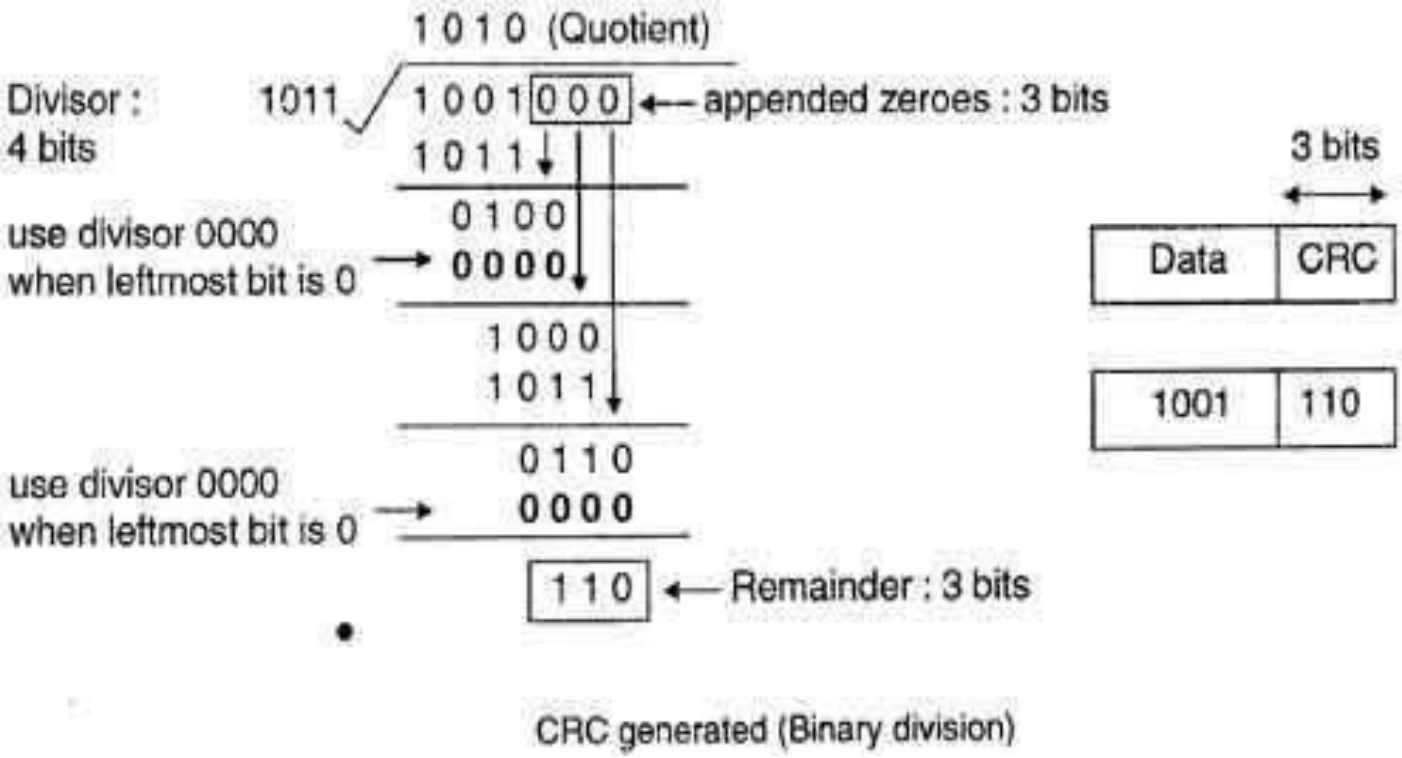
```
1 0 1 0 0 0 0 0 0 0
        + 0 1 1
-----------------
1 0 1 0 0 0 0 0 1 1
```

```
1 0 0 1 | 1 0 1 0 0 0 0 0 1 1
        @ 1 0 0 1
          ---------
          0 0 1 1 0 0 0 0 1 1
            @ 1 0 0 1
              ---------
              0 1 0 1 0 0 1 1
                @ 1 0 0 1
                  ---------
                  0 0 1 1 0 1 1
                    @ 1 0 0 1
                      ---------
                      0 1 0 0 1
                        @ 1 0 0 1
                          ---------
                          0 0 0 0
```

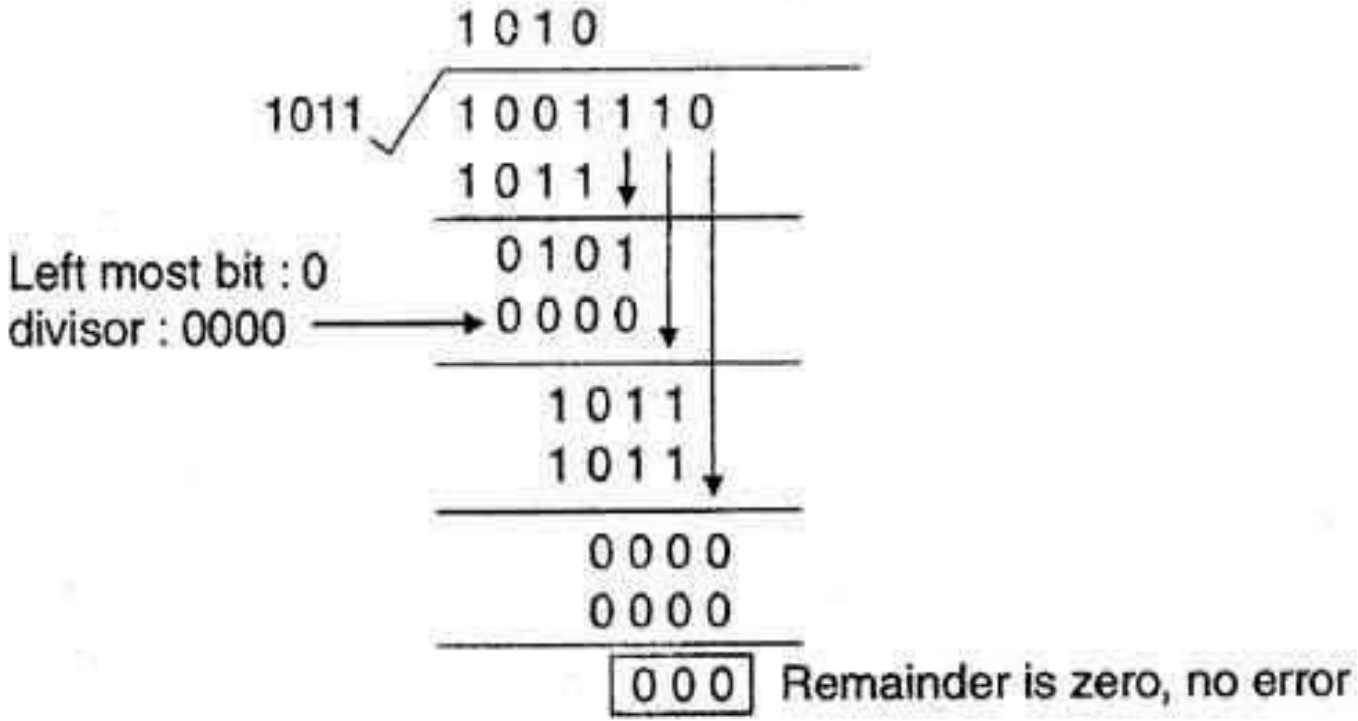
Receiver

Zero means data is accepted

Example-3 Data unit 1001000 is divided by 1011.



CRC generated (Binary division)



CRC decoded (binary division)

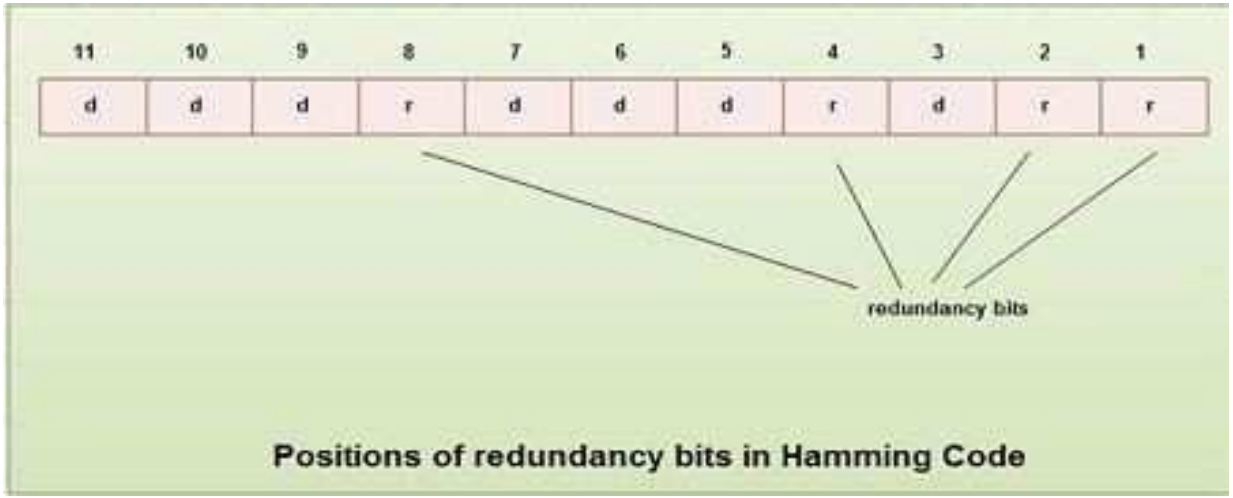
Checksum

- In this error detection scheme, the following procedure is applied
- **Sender-**
- 1.Divide the data into x segment and also every section is of Y bits.
- 2.Perform binary addition followed by 1’s complement ,which shall be the checksum.
- 3.Send the data after adding the checksum in it.
- **Receiver-**
- 1. Divide the received data into X section and every section of Y bits.
- 2.Perform binary addition followed by 1’s complement
- If result is all zero then there is not any error or else error is detected.

Hamming Code

- Hamming code is technique developed by R.W. Hamming for error correction. This method corrects the error by finding the state at which the error has occurred.
- **Determining the positions of redundancy bits**
- Till now, we know the exact number of redundancy bits required to be embedded with the particular data unit.
- We know that to detect errors in a 7 bit code, 4 redundant bits are required.
- Now, the next task is to determine the positions at which these redundancy bits will be placed within the data unit.
  - These redundancy bits are placed at the positions which correspond to the power of 2.
  - For example in case of 7 bit data, 4 redundancy bits are required, so making total number of bits as 11. The redundancy bits are placed in position 1, 2, 4 and 8 as shown in fig.

Hamming Code



Positions of redundancy bits in Hamming Code

Thank You !